

Ko'p o'zgaruvchi murakkab funksiyaning differensiallanuvchiligi

1^o. Murakkab funksiyaning differensialli. Differensial shaklning invariantligi.

Ma'lumki, $x_i = \varphi_i(t_1, \dots, t_k)$ funksiyalar ($i = 1, 2, \dots, m$) $t^0 = (t_1^0, \dots, t_k^0)$ nuqtada differensiallanuvchi bo'lib, $f(x) = f(x_1, x_2, \dots, x_m)$ funksiya mos $x^0 = (x_1^0, x_2^0, \dots, x_m^0)$ nuqtada ($x_1^0 = \varphi_1(t^0), x_2^0 = \varphi_2(t^0), \dots, x_m^0 = \varphi_m(t^0)$) differensiallanuvchi bo'lsa, murakkab funksiya $t^0 = (t_1^0, \dots, t_k^0)$ nuqtada differensiallanuvchi bo'ladi.

Modomiki, $f(x(t))$ funksiya $t_1, t_2, \dots, t_k$ o'zgaruvchilarga bog'liq ekan, unda

$$df = \frac{\partial f}{\partial t_1} dt_1 + \frac{\partial f}{\partial t_2} dt_2 + \dots + \frac{\partial f}{\partial t_m} dt_m \quad (3)$$

bo'ladi.

Murakkab funksiyaning xususiy hosilalarini hisoblash formulalaridan foydalanib topamiz:

$$\frac{\partial f}{\partial t_1} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_1} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_1} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_1},$$

$$\frac{\partial f}{\partial t_2} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_2} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_2} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_2},$$

.....

$$\frac{\partial f}{\partial t_k} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t_k} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t_k} + \dots + \frac{\partial f}{\partial x_m} \frac{\partial x_m}{\partial t_k}.$$

Bu xususiy hosilalarni (3) ifodadagi $\frac{\partial f}{\partial t_1}, \frac{\partial f}{\partial t_2}, \dots, \frac{\partial f}{\partial t_k}$ larning o'rniga qo'yamiz.

Natijada

funksiyalari $E \subset R^m$ to'plamda berilgan bo'lib, $x^\circ = (x_1^\circ, x_2^\circ, \dots, x_m^\circ) \in E$ nuqtada differensiallanuvchi bo'lsin. U holda:

$$1) d(u+v) = du + dv,$$

$$2) d(u \cdot v) = vdu + u dv,$$

$$3) d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2} \quad (v \neq 0)$$

bo'ladi.

Bu munosabatlardan birini, masalan, 3) ning isbotini keltiramiz.

◀ Aytaylik,

$$F = \frac{u}{v}$$

bo'lsin. Bu holda F funksiya u va v larga va u va v lar o'z navbatida $x_1, x_2, \dots, x_m$ o'zgaruvchilarga bog'liq bo'lib, murakkab funksiya ega bo'lamiz. Differensial shaklning invariantli xossasiga ko'ra

$$dF = \frac{\partial F}{\partial u} du + \frac{\partial F}{\partial v} dv$$

bo'ladi. Ravshanki,

$$\frac{\partial F}{\partial u} = \frac{1}{v}, \quad \frac{\partial F}{\partial v} = -\frac{u}{v^2}$$

Demak,

$$dF = \frac{1}{v} du - \frac{u}{v^2} dv = \frac{vdu - u dv}{v^2},$$

ya'ni

$$d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2}$$

bo'ladi. ▶

Mashqlar

1. Ushbu

$$f\left(x^2 + y^2, \operatorname{arctg} \frac{y}{x}\right) \quad (x^2 + y^2 > 0)$$

funksiyaning differensialli topilsin.

2. Ushbu

$$\alpha = (1, 02)^{3,01}$$

miqdorning taqribiy qiymati topilsin.

Quyidagi funksiyalarning xususiy hosilalarini toping

$$3. f(x, y) = \frac{x - y}{x + y}$$

$$4. f(x, y) = \frac{\cos x}{\cos y}$$

$$5. f(x, y) = (\sin x)^{\cos y}$$

Quyidagi murakkab funksiyalarning xususiy hosilalarini toping

$$6. f(x, y) = x^2 y^3 \quad x = t \quad y = t^3$$

$$7. f(x, y) = F \quad x = u^2 + v^2 \quad y = uv$$

$$8. f(x, y) = x^y \quad x = \sin u \quad y = \cos v$$

Quyidagi funksiyaning diffrensiyalini toping

$$9. f(x, y) = \arctg \frac{y}{x} + \arctg \frac{x}{y}$$

$$10. f(x, y) = \left(\frac{x}{y} \right)^y$$